

Hackathon Day 1

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AI4Imaging Summer School

7 - 11 September 2026



MEDICAL UNIVERSITY
OF VIENNA

CIR

Hackathon Day 1
AI4Imaging Summer School

Getting Started!

3. open the terminal

4. clone our Hackathon git repository:

```
git clone https://tinyurl.com/SummerSchoolMedUni
```

5. access the folder:

```
cd SummerSchoolMedUni
```

6. install the required packages via terminal:

```
pip install --user -r requirements.txt
```

7. you can now start coding by using the notebook

day1_pneumonia_pipeline.ipynb as a basis.

Tutors



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Hackathon Overview

 Monday - Task #1 PneumoniaMNIST

 Tuesday - Task #2 ISIC

 Wednesday - finish up tasks & prepare presentation. **Download your work**

 Friday - Final presentation



You won't be able to access your AWS account after Wednesday noon (12 pm)

1 Data Loading and Exploration



Selecting and initializing the dataset

- Load the "PneumoniaMNIST" dataset from medmnist
- Apply preprocessing transformations from Pytorch



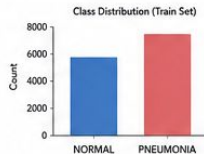
Visualizing the dataset

- Print out dataset properties
- Visualize some example instances



Plotting class distributions

- Count the number of labels for each data subset
- Visualize in a plot



Dataloader

- Setup the dataloader



Data augmentation *optional*

- Investigate on data augmentation methods
- Generic augmentations with torchvision
- Specific augmentations with monai

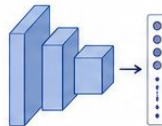


2 Model Training and Evaluation



Implement classification model

- Define training and model parameters (learning rate, optimizer, epochs, ...)
- Define classification model (see e.g. MONAI networks)
- Vary the training and model parameters and document your observations regarding model performance, training time, etc.



Define performance metrics

- Which metrics are suitable for evaluating your classification model? (see e.g. scikit-learn)
- What exactly do the individual metrics mean? How can you interpret the results?

Examples of Metrics

Accuracy
Precision
Recall (Sensitivity)
Specificity
F1-Score
ROC-AUC
...



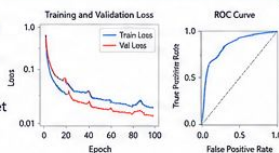
Training loop

- Iteratively pass batches through your network, calculate loss and update model weights
- Evaluate model performance on your validation data
- Save your model



Final model evaluation

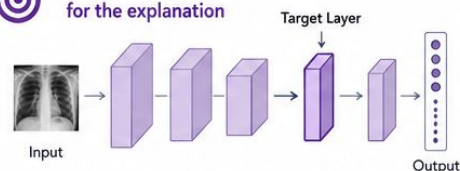
- Visualize model performance (loss, evaluation metrics)
- Model inference on the test set



3 Explainable AI



Define the target layer for the explanation



Apply explainability methods

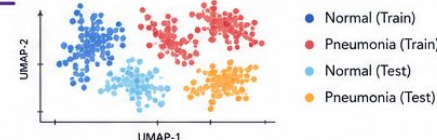
- Grad-CAM / Score-CAM
- Integrated Gradients
- Guided Backpropagation
- Occlusion Sensitivity
- ...



Visualize as heat maps



Plot latent space features *optional*

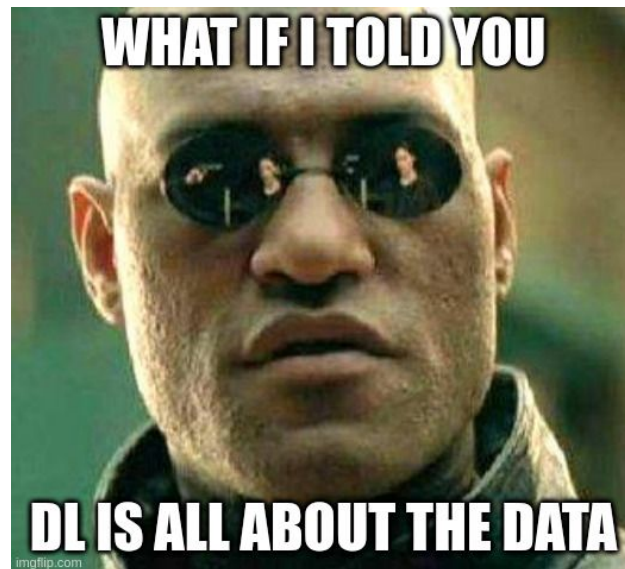


Data Loading and Exploration

Task 1

Data Loading and Exploration

- The main goal of this session is learning about:
 - loading datasets
 - preprocessing datasets
 - data augmentation
 - exploring of a dataset by visualization



Tasks

1. Selecting and initializing the dataset

- Load the “PneumoniaMNIST” dataset from medmnist
- Apply preprocessing transformations from Pytorch

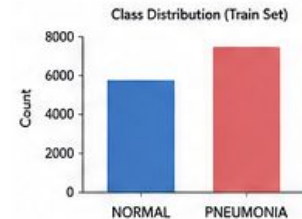


2. Visualizing the dataset

- Print out dataset properties
- Visualize some example instances

3. Plotting class distributions

- Count the number of labels for each data subset
- Visualize in a plot



Tasks

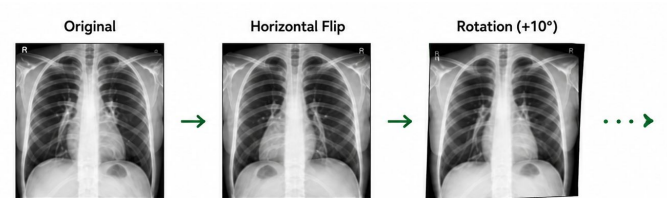
4. Dataloader

- Setup the dataloader

5. Data augmentation **optional**

- Investigate on data augmentation methods
- Generic augmentations with torchvision

Specific augmentations with monai



6. ChestMNIST **optional**

- Investigate a multi-class problem
- Advanced visualizations and evaluations

Training and Evaluation

Task 2

Training and Evaluation

- The main goal of this session is:
Train and **evaluate** a classification model for detection of pneumonia
- In the process:
 - Define training and model parameters
 - Implement a training loop
 - Evaluate and visualize model performance

Tasks

1. Implement classification model:

- Define training and model parameters (learning rate, optimizer, epochs, ...)
- Define classification model (see e.g. <https://docs.monai.io/en/stable/networks.html#nets>)
- Vary the training and model parameters and document your observations regarding model performance, training time, etc.

2. Define performance metrics

- Which metrics are suitable for evaluating your classification model? (see e.g. https://scikit-learn.org/stable/modules/model_evaluation.html)
- What exactly do the individual metrics mean? How can you interpret the results?

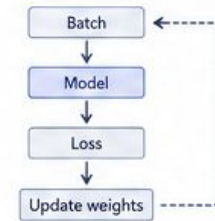
Examples of Metrics

Accuracy
Precision
Recall (Sensitivity)
Specificity
F1-Score
ROC-AUC
...

Tasks

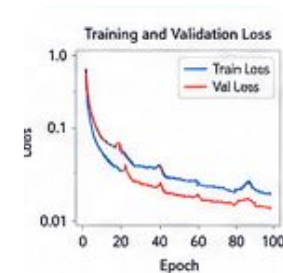
3. Training loop

- Iteratively pass batches through your network, calculate loss and update model weights
- Evaluate model performance on your validation data
- Save your model



4. Final model evaluation

- Visualize model performance (loss, evaluation metrics)
- Model inference on the test set



Explainable AI

Task 3

Learning Outcome

The main goal for this session is learning about:

- How to access a specific layer in a model
- Applying XAI methods
- Visualizing in the form of heat maps

Explainable AI

- AI models are often “black-box” algorithm
- Interpretation is difficult even for experts



Explainable AI

Goals of Explainable AI:

- Build up trust in AI Models
- Finding mistakes and biases of Algorithms
- Knowledge extraction from black-box models

GradCAM

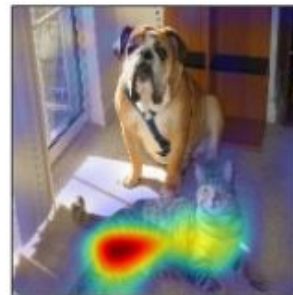
- Grad-CAM helps us understand what a model “sees” in an image
- Here a model for object classification is analyzed
- Grad-CAM helps to visualize the contributing factors for “cat” and “dog”



Original Image



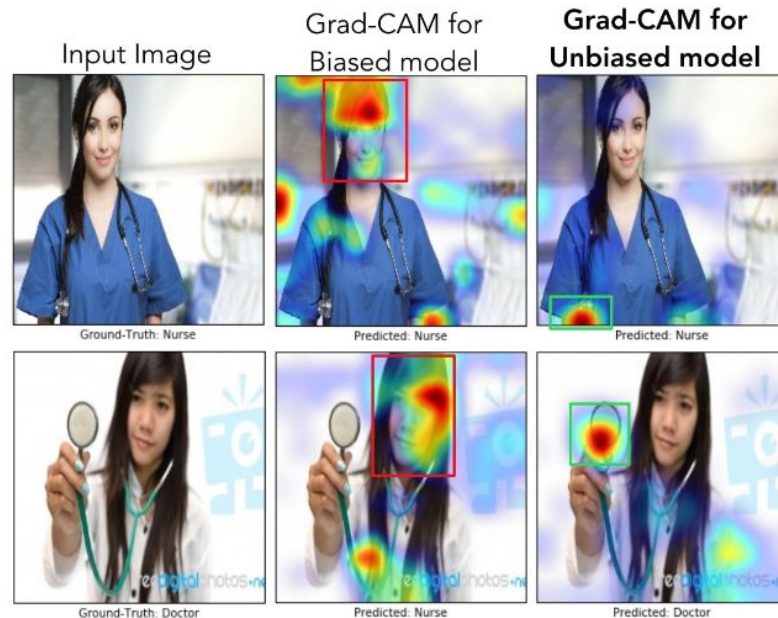
Grad-CAM “Dog”



Grad-CAM “Cat”

GradCAM

- A model for object classification is analyzed
- Grad-CAM helps to visualize the bias of a model
- The biased model focuses on the face of the nurse or doctor. Therefore, giving a wrong prediction



Tasks

Continuing on the Session 3 of the notebook: `model_interpretation_baseline.ipynb`

1. Load the dataset and your model
2. Define the target layer for the explanation
3. Apply explainability methods
4. Visualize as heat maps
5. Plot latent space features optional

Optional Tasks

Task 1: Multi-label classification

- Use the ChestMNIST dataset from MedMNIST
- Transfer the code to solve a multi-label problem
- How can you handle class imbalances?

Task 2: Large Language Models (LLMs)

- Use LLMs to generate radiological reports for the PneumoniaMNIST dataset
- See <https://huggingface.co/microsoft/Phi-3-vision-128k-instruct>

Task 3: Generative data augmentation

- Train a Generative Adversarial Network (GAN) to enlarge your dataset with new generated healthy and pneumonia images
- Consider to use a pre-trained GAN (see https://github.com/Project-MONAI/tutorials/blob/main/modules/mednist_GAN_tutorial.ipynb)

Happy coding !

Feel free to ask any questions!